

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2023-24

Paper No. MTB-AM-403: Ancillary Mathematics II

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 077613

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Give a relation and a function from $A = \{1, 2, 3\}$ to $B = \{x, y, z\}$.
- (b) Define the domain and range of a function. Find the domain for which $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ are equal.
- (c) Evaluate $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x - 2}$.
- (d) Show that $f(x) = 7x - 3$ is increasing on $\mathbb{R}$.
- (e) Evaluate $\int \frac{1}{\sqrt{x}} dx$.
- (f) Find $\int x \cos x dx$.
- (g) Determine the order and degree of $(y'')^2 + (y')^3 + (y)^4 + y^5 = 0$.

Question 2.

- (a) Define the derivative of a real-valued function. Evaluate the derivative of $f(x) = \sin x$ using the definition. [7]
- (b) Find the equations of the tangent line and normal to the curve $y = x^2 + 3x + 2$ at the point $(1, 6)$. Explain its geometrical significance. [7]

Question 3.

- (a) For any positive integer n , prove that $\lim_{x \rightarrow a} x^n = a^n$. [7]
- (b) What is the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$? Hence evaluate $\lim_{x \rightarrow 0} \frac{a \sin x}{x}$. [7]

Question 4.

- (a) The volume of a cube is increasing at $9 \text{ cm}^3/\text{s}$. How fast is the surface area increasing when the edge length is 10 cm ? [7]
- (b) Find the intervals on which $f(x) = \sin x + \cos x$, $0 < x < 2\pi$, is increasing or decreasing. [7]
-

Question 5.

- (a) Find local maximum and minimum values of $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$. [7]
- (b) A manufacturer can sell x items at price $\left(5 - \frac{x}{100}\right)$ rupees each. The cost price of x items is $\left(\frac{x}{5} + 500\right)$ rupees. Find the number of items to sell for maximum profit. [7]
-

Question 6.

- (a) Show that $\int \sec x \, dx = \log |\sec x + \tan x| + C$. [7]
- (b) Find $\int \frac{3x - 2}{x^2 + x - 2} \, dx$. [7]
-

Question 7.

- (a) In a bank, principal increases continuously at 5% per year. In how many years does Rs. 1000 double itself? [7]
- (b) Show that $(x - y)\frac{dy}{dx} = x + 2y$ is homogeneous, and solve it. [7]
-